

Quad Packing

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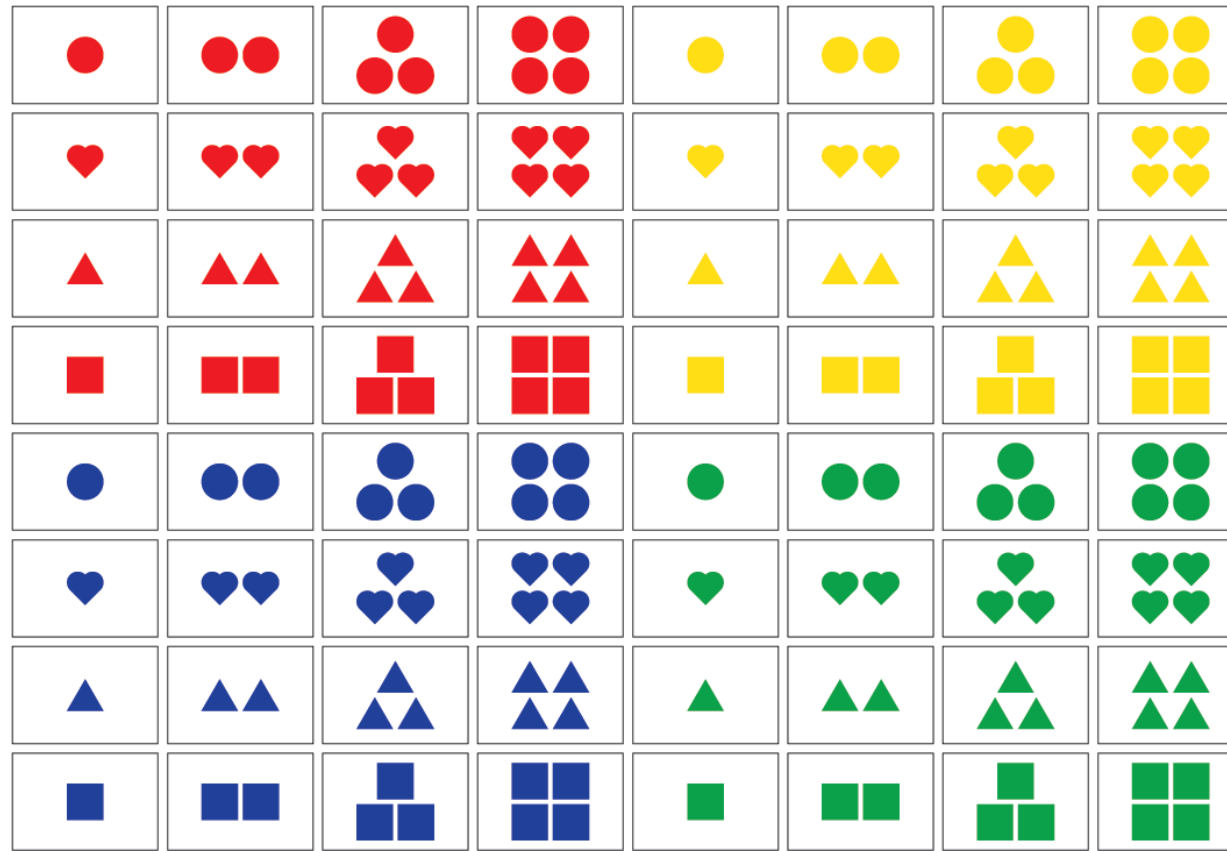
Acknowledgements

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Outline

- I. What is Quads?
- II. Fundamental Theorem of Quads
- III. Vector Representation & Geometry
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What is Quads?



What makes a quad?

Expressions of attributes are all the same, all different, or in pairs.



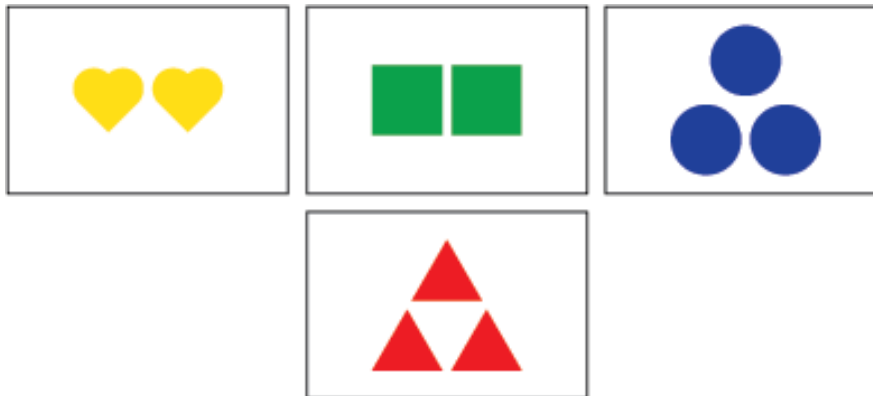
Number is all the same, shape is in pairs, and color is all different

Fundamental Theorem of Quads

For any three cards, there exists a unique fourth that completes a quad.



What card completes the quad?



Vector Representation & Geometry

We can assign vectors in $\mathbb{Z}_2^6$ to each card in quads by using the following table.

(0,0)	1	(0,0)	circle	(0,0)	red
(0,1)	2	(0,1)	heart	(0,1)	yellow
(1,0)	3	(1,0)	triangle	(1,0)	green
(1,1)	4	(1,1)	square	(1,1)	blue

Ex.  = (0,0,1,1,1,0)

Theorem: Cards a, b, c, d form a quad if and only if $a + b + c + d = 0$.

Corollary: Given a, b, c , the value $a + b + c$ completes the quad.

Quad Packing Problem

What is the maximum number of quads given k cards in Quad-64?

Alternatively: What is the maximum number of planes given k points in $\mathbb{Z}_2^6$?

Let $q(k, d) =$ maximum number of quads given k cards in $\mathbb{Z}_2^d$.

Small k

$$q(4, d) = 1$$

- *Trivial.*

$$q(5, d) = 1$$

- *Proof:* Let a, b, c, d, e be our cards.
- Some subset of 4 cards form a quad, otherwise there are no quads.
- WLOG $a + b + c + d = 0$.
- Any other quads must contain e and 3 of a, b, c, d .
- By FToQ, any 3 of a, b, c, d require the fourth to form a quad.
- Therefore, $q(5, d) = 1$.



4 cards with 1 quad

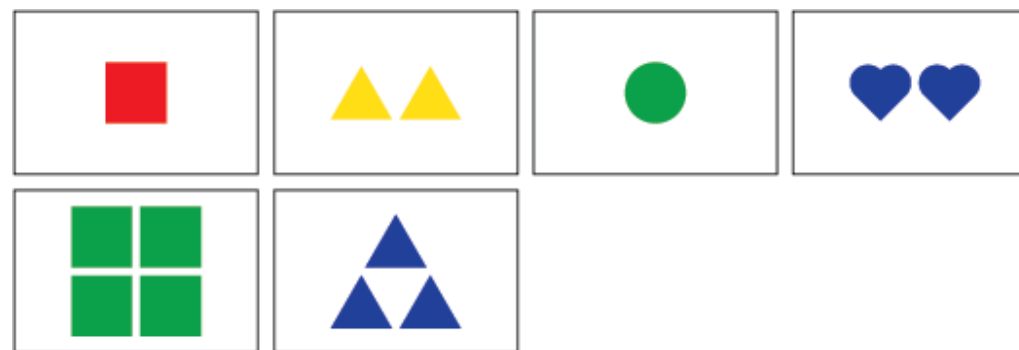


5 cards with 1 quad

Small k

$$q(6, d) = 3$$

- *Proof:* Let a, b, c, d, e, f be our cards.
- Some subset of 4 cards form a quad, otherwise there are no quads.
- WLOG $a + b + c + d = 0$.
- Any new quad cannot contain 3 of a, b, c, d , so it must contain e, f .
- WLOG $a + b + e + f = 0$. We then also have $c + d + e + f = 0$.
- Any combination of e, f with one of a, b, c, d defines a pre-existing quad.
- Therefore, there are no more possible quads and $q(6, d) = 3$.



6 cards with 3 quads

Bonus: See if you can find all three!

$$q(7, d) = 7$$

- *See final paper for proof!*

An Upper Bound

Statement: For any k such that $4 \leq k \leq 2^d$,
 $q(k, d) \leq \frac{\binom{k}{3}}{4}$.

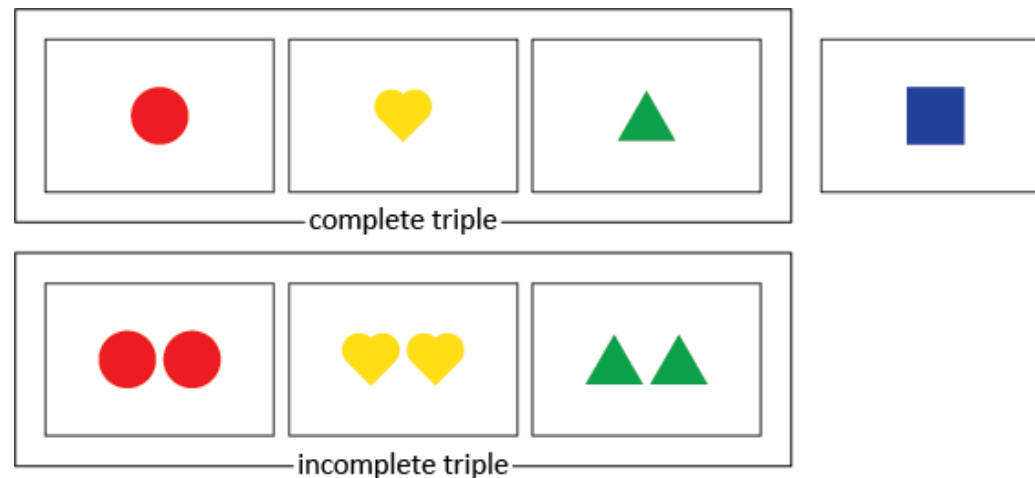
◦ *Proof:*

- Fix a maximal set of k cards.
- Any set of three cards forms a triple.
- Let a be the number of complete triples and b be the number of incomplete triples.

◦ Then:

$$\frac{\binom{k}{3}}{4} = \frac{a}{4} + \frac{b}{4} = \frac{4q(k, d)}{4} + \frac{b}{4} = q(k, d) + \frac{b}{4}$$

$$q(k, d) \leq \frac{\binom{k}{3}}{4}$$



Powers of 2

Statement: For an integer n such that $2 \leq n \leq d$, $q(2^n, d) = \frac{\binom{2^n}{3}}{4}$.

◦ *Proof:*

- Let $x_1, x_2, \dots, x_n$ be distinct linearly independent non-zero vectors in $\mathbb{Z}_2^d$.
- $|\text{span}\{x_1, x_2, \dots, x_n\}| = 2^n$.
- Let $a, b, c \in \text{span}\{x_1, x_2, \dots, x_n\}$. Then $a + b + c \in \text{span}\{x_1, x_2, \dots, x_n\}$.
- All triples are therefore complete, so this set has $\frac{\binom{2^n}{3}}{4}$ quads.
- As $q(2^n, d) \leq \frac{\binom{2^n}{3}}{4}$, we must have $q(2^n, d) = \frac{\binom{2^n}{3}}{4}$.

QuadFinder Code

k	d	$q(k,d)$
4	6	1
5	6	1
6	6	3
7	6	7
8	6	14
9	6	14
10	4	18

```
void iterate(int *a, int i, int size, int n, ll &count) {
    int start = 0;
    if (i > 0) {
        start = a[i-1]+1;
    }
    for (a[i] = start; a[i] < size; a[i]++) {
        if (i == n-1) {
            vector<int> subset;
            for (int k = 0; k < n; k++) {
                subset.push_back(a[k]);
            }
            ll curcount = 0;
            for (int i = 0; i < n; i++) {
                for (int j = i+1; j < n; j++) {
                    for (int k = j+1; k < n; k++) {
                        for (int l = k+1; l < n; l++) {
                            int cur = 0;
                            cur ^= subset[i];
                            cur ^= subset[j];
                            cur ^= subset[k];
                            cur ^= subset[l];
                            if (cur == 0) {
                                curcount++;
                            }
                        }
                    }
                }
            }
            count = max(count, curcount);
        }
        else {
            iterate(a, i+1, size, n, count);
        }
    }
}
```

QuadFinder Iterate Function

Conjectures

For $k \leq 2^d$, $q(k, d) = q(k, d + 1)$.

$$q(2^n + 1, d) = q(2^n, d).$$

$$q(2^n - 1, d) = q(2^n, d) - \frac{\binom{2^n - 1}{2}}{3}.$$

Let S_k be a (k, d) -maximal set of k cards. $\exists a \in \mathbb{Z}_2^d$ such that $S_k \cup \{a\}$ is $(k + 1, d)$ -maximal.

Based on our results from QuadFinder for small k , the first three conjectures appear to hold.

The last conjecture was proposed as this is how we were able to find results for $4 \leq k \leq 7$.

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